Daily Drilling Report
SPD19B-03
8/29/2014

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
33.0	12 1/4"	178.00	72.83	31.70	94.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
13 3/8" Shoe, 1,487.00mKB	225,147.73	41,742.16	7,657,151.52		28.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
2,437.00	2,496.00		59.00	13.10	4.5

Operations at Report Time
Drilling @ 2506m.

Operations Summary
Con't RIH directional BHA#9 on 5 7/8" DP to BTM at 2437 m. Resume Dir. drilling 12 1/4" hole section f/ 2437m to 2496m.

Operations next 24hrs
Cont' drilling 12 1/4" hole section.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Jabbar Mohammadi	Night Drilling Sup.
Behnam Jahangiri/Hamed Daneshkhah	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/23/2014	10/13/2014
BOP Test	8/20/2014	9/11/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/23/2014	10/13/2014
H2S Drill	8/16/2014	9/16/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	4/16/2014	6/16/2014
Pit Drill	8/25/2014	10/10/2014
Trip Drill	8/28/2014	10/18/2014

STOP Cards

Number of Reports: 40

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	08:00	8.00	12 1/4"	D	04	P	Cont' RIH directional BHA#9 on 5 7/8" DP f/231m to 2413m. W/R down f/ 2413m to 2437m (Bottom) w/ 800 gpm/ 45 rpm.
08:00	23:00	15....	12 1/4"	D	03	P	Fan Btm and profile new bit. Cont' directional drilling 12 1/4" hole in rotation mode f/ 2437m to 2496m w/750-810 gpm/ 1650-2050 psi/ 5-40 kips/ 35-60 rpm/ 14-22 kft-lbs. ** Drilling with NL/NG. ** 20 bbl Hi-Vis pill when STD down. ** B/R and ream down each stand drilled once. ** Take survey before connection.
23:00	00:00	1.00	12 1/4"	D	07	N	While drilling in Massive Anh. observed changes in mud properties (decrease in mud weight f/11.4 to 11.3 ppg, decrease in viscosity f/51 to 48 sec/qt, increase in hardness f/550 to 1000 ppm). Suspect formation water influx. Decided to stop drilling & treat the mud before entering S8. Start increasing and conditioning the mud. Meanwhile reciprocate the string w/ 30 rpm/ 700 gpm.
00:00	00:00		12 1/4"				Aug 30, 2014 00:00-03:00 Con't increasing MW to 11.6 ppg and treating the mud. Meanwhile keep the string in motion w/ 30 rpm / 700 gpm. Bring the mud properties to the normal range. 03:00-06:00 Resume drilling f/2496m 2506m w/ 800 gpm/ 2150 psi/ 27-37 kips/ 60 rpm/ 15-22 kft-lbs >>Drilling Practice in "S8" Member (Geologist Expectation of "S8" @ ±2498m): ** For the first 10m in "S8", back ream gently to top of S8 (2498m) every 1m drilled. ** Last Sample @ 2500m: Anhy (60%) + Clst (20%) + Sh (20%).

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PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
2,496.00	KCl-Polymer Mud	11.30	48	19.0	23.000	4.5
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
9.5	0.600	1.200	1.400			
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
5	8	2.0	80.0	18.0	55,000.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
7.000	12.000	14.000	2.4/97.6	63.0	2019.0	1725.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
5076.0	136.0	0.0	0.01		12.30579	0.448

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
1.00	6.	135	12.0	998.0	34.5

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Sea Command	Supply Boat	8/27/2014 20:50	8/29/2014 15:30	B/L Cargo for Kish. Dep for Sahar 1 at 1530 Hrs.
Morshed	Supply Boat	8/29/2014 19:30	8/30/2014 02:40	Delivered Food Container & Cargo, Embarked 02 PAX, departed for Sahar-2
Forte	Supply Boat	8/29/2014 19:30		Disembarked 02 PAX f/ Sahar-2, Delivered all Cargo. Delivering Cement in progress.
Marina-1	Supply Boat	8/16/2014 15:15		Standby at Anchorage.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Barite Bulk	17.0	249.0	190.0
Caustic Soda 2014	21.0	340.0	207.0
Glydril	40.0	100.0	56.0
KCL	4.0	52.0	28.0
Starch	160.0	1,080.0	434.0
Pac-LV		422.0	175.0
Soda Ash 25 Kg	46.0	550.0	309.0
PTS (Temp. Stabilizer)	1.0	19.0	18.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Potable Water	20.0	14.0	102.0
Drill Water	7.0		323.0
Diesel Fuel	11.0		172.0
Cement G	0.0	0.0	132.0
Bentonite Bulk	0.0	0.0	8.0
Bentonite BB	0.0	0.0	0.0
Barite BB	0.0	0.0	0.0
Barite Bulk	17.0		190.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
9	12 1/4	Varel	VM616PGUX-V2	6011971	PDC	___	12/12/12/12/12/12/12/12/12
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.99	13.10	113,283.60	2,437.00	2,496.00	59.00	4.5	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
750	810	1,650.0	2,050.0	7	40	125	157
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
16.34	13.10	100.00	0.00	382	343	296	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
2,437.00	2,438.30	0.38	21,000.0	750	35	1,650.0	7	343
2,438.30	2,473.30	9.78	21,000.0	800	45	1,900.0	40	343
2,473.30	2,496.00	2.94	22,000.0	810	60	2,050.0	40	343

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
3 jts 5 7/8" HWDP	5 7/8	4.000	27.83	231.39
7" Drilling Jar (COR-3022-AD)	7	2.938	5.85	203.56
16 jts 5 7/8" HWDP	5 7/8	4.000	149.23	197.71
X/O	7	2.813	0.79	48.48
1 jt 8 1/4" DC	8 1/4	2.810	9.46	47.69
8 1/2" Float Sub	8 1/2		0.79	38.23
8" NMDC	8	3.500	9.44	37.44
8" TM - Sub	8		2.03	28.00
8" NMDC	8	3.500	9.46	25.97
11 7/8" S. Stabilizer (COR3103AD)	11 7/8	2.813	2.16	16.51
8" PNMDC	8 1/16	3.500	4.03	14.35
9 5/8" Motor (AKO: 1.00° , Sleeve: 12 1/8")	9 5/8		9.95	10.32
12 1/4" PDC bit	12 1/4		0.37	0.37

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

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Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.35	1,200.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,200.0	7.7	1,200.00		14.03	3,413.7

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
2,443.10	247.50	20.50	2,316.66	-303.74	-654.38	721.44	0.39	-0.32
2,472.18	247.90	20.70	2,343.88	-307.63	-663.85	731.66	0.25	0.21

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
Neyriz		2,240.00	2,128.14
Up Dashtak		2,266.00	2,152.17
Evaporite "B"		2,338.00	2,218.78
Upper Shale		2,422.00	2,296.88
Massive Anh.		2,473.60	

General Remarks

Comment

>>> Requirements: (70 BB CaCo3 f/ MI MR #13).

>>> Sent:
** Sea Command to Kish: (PPL basket containing casing protectors) + (Metal skip containing casing slings & clamps) + (Garbage)

>>> Received:
** Forte f/ Kish: (Mehran CMT Additives) + (10 jts of 10 3/4" Tenaris Blue Casing)

** Morshed f/ Sharjah: (ABAN food provision) + (400 Sxs starch) + (96 x cans Caustic Soda) + (200 Sxs Soda Ash) + (8 x drums PTS 200) + (40 x drums Glydrill MC) + (32 x cans MI CIDE) + (160 Sxs PAC LV) + (HPOGC: 1 x 9 5/8" Rotor 6 lobes + 4 x lithium batteries)